

Dr. Shirshendu Das

Assistant Professor
Department of CSE
Indian Institute of Technology Hyderabad
Telangana, India, 502284

+91-9435277250
shirshendu.3it@gmail.com
shirshendu@cse.iith.ac.in
<https://sites.google.com/view/shirshendudas/home>
i10-index: 12, Avg. IF: 2.85 Citation: 355

AREA OF SPECIALIZATION

My area of expertise is in **Computer Architecture**. The two principal subareas in which I am now working are:

- **Hardware Security** - Security issues in Last Level Cache (LLC), DRAM-based main memories, NVM-based cache memories, and Network on Chip (NoC). I am exploring new security concerns for modern computing devices and also developing effective countermeasures.
- **Performance Enhancement and Energy Reduction of Memory Devices** - Reducing refresh overhead of DRAM-based main memory, Reducing the overhead of write operation in NVMs, Process variation aware cache/main memory reconfiguration, Replacement policy, Cache partitioning, and Cache prefetching.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
PhD (CSE)**	Indian Institute of Technology Guwahati, Assam	N/A	Dec 2010- Jan 2016
MTech. (CSE)	Indian Institute of Technology Guwahati, Assam	9.29	Aug 2008- Jun 2010
MSc. (CS)	Assam University Silchar, Assam	73.5%	Jul 2005- Jun 2007
BSc. (CS)	Assam University Silchar, Assam	66%	Jul 2002- Jun 2005

****PhD Supervisor:** Prof. Hemangee K. Kapoor, Department of CSE, IIT Guwahati.

****Thesis Title:** Effective Utilization of LLCs by Managing Associativity, Placement and Mapping

EXPERIENCE

- **Indian Institute of Information Technology Guwahati (IIIT Guwahati)** *Aug 2015 - Jun 2018*
Assistant Professor
Guwahati, Assam
– I have joined IIIT Guwahati as an Assistant Professor just after submitting my PhD thesis. My PhD degree was awarded in January 2016.
- **Indian Institute of Technology Ropar (IIT Ropar)** *Jun 2018 - Nov 2022*
Assistant Professor
Punjab, India
- **Indian Institute of Technology Hyderabad (IIT Hyderabad)** *Dec 2022 - Till Date*
Assistant Professor
Telangana, India
– I am currently working as Assistant Professor Grade-I with Pay Level 13A1.

PROJECTS UNDERTAKEN

- **Efficient Utilization and Refresh Overhead Minimization of eDRAM based Last Level Cache** - SERB Funded Project. **Completed in March 2023**. Total: 35L. (*Started at IIT Ropar and Completed at IITH.*)
- **Reducing Energy Consumption and Operating Temperature of Last Level DRAM Cache in Multicore Systems** - SERB Funded Project. **Completed in March 2021**. Total: 15L.
- **Next Gen Computer Architecture** - Intel India, **Completed in June 2025**, Total: 24L
- **Efficient Use of STTRAM for Designing Secure Last Level Cache of Modern Chipmultiprocessors** - IIT Hyderabad Seed Grant, **Ongoing**, Total: 30L
- **Do not Forget Cache Content Management Techniques While Designing Secure Last-level Cache** - ANRF Funded Project (CRG), **Ongoing**, Total: 23L
- **EPOCH: Exploring Potential of Multi-Level Cell and Multi-Retention MRAM Based Caches for Heterogeneous Cores** - ANRF Funded Project (SERB ARG), **Under review**, Total: 107L

MANPOWER ASSOCIATED WITH ME: (INCLUDING MY PREVIOUS INSTITUTE)

- **PhD students:** 06 – ongoing 04, **awarded (IIT Ropar) 02**, ongoing at IIT Hyderabad 02.
- **M.Tech students:** 03 (ongoing at IITH), 03 (completed at IITH), 17 (Completed from IIT Ropar).
- **B.Tech students:** 02 (Ongoing at IITH), 05 (Completed at IITH), 11 (Completed at IIT Ropar)

COURSES TAUGHT (INCLUDING MY PREVIOUS INSTITUTES)

Software Development Fundamentals, Database Management Systems, Computer Architecture, Introduction to Programming and DS, Data Structures, Memory Systems and Architecture, Computer Systems, Advanced Computer Architecture, Compilers, and TOC.

ACHIEVEMENTS

- **TCS Research Fellowship Award** - Received the fellowship during my PhD 2011 - 2015
- **Additional top up salary** - Selected for the additional top up salary (20K per month) at IIT Hyderabad, given to the exceptional faculty candidates for one year. Dec 2022
- **M.Tech best Thesis Award by My Student** - One of my M.Tech student got Institute best thesis award. 2021
- **Reviewer of journals/transactions** - Reviewed articles for IEEE TPDS, IEEE TCAD, Elsevier VLSI, Elsevier MICPRO, Elsevier JSA, etc.

PUBLICATION STATISTICS

- **Journals:** Average IF: 2.85 , Total Published: 19 ([7 Q1](#), 12 Q2), Total under submission: 03. Some important places where I have published are IEEE TPDS, ACM TACO, ACM TODAES, Elsevier JPDC, Elsevier JSA, Elsevier MICPRO, etc.
- **Conferences:** Total published: 31 ([7 A-Rank](#)). Total under submission: 02. Some important venues where I have published are: HOST 2025, DATE 2021/2024, ISQED 2021/2019, ACM-SAC 2016/2015, ISVLSI 2017, and VLSID 15/17, etc.

PUBLICATIONS

Journals (After Joining IITH):

1. Josna Philomina, Rekha K James, Palash Das, **Shirshendu Das** and Daleesha M Vishwanathan, "Exploiting Virtual Channel Allocation Policies in STT-RAM Buffers of NoC Routers Through Hardware Trojan" *Elsevier Journal of System Architecture, Accepted (JSA)*, October 2025. **IF: 4.1, Q1**
2. Prabuddha Sinha, Krishna Pratik Bv, **Shirshendu Das** and Venkata Kalyan Tavva, "SmartDeCoup: Decoupling the STT-RAM LLC for even Write Distribution and Lifetime Improvement " *Elsevier Journal of System Architecture, Accepted (JSA)*, February 2025. **IF: 4.1, Q1**
3. Prabuddha Sinha, Mangena Likhith Sai, **Shirshendu Das**, and TV Kalyan, "TENDRA: Targeted Endurance Attack on STT-RAM LLC", *IEEE Embedded Systems Letters (ESL)*, Accepted, Nov 2024. **IF: 2, Q2**
4. Atul Kumar, **Shirshendu Das** and Basant Subba, "HTree: Hardware Trojan Attack on Cache Resizing Policies", *IEEE Embedded Systems Letters (ESL)*, 16(3), 2023, p. 263-266. **IF: 2, Q2**
5. Jaspinder Kaur and **Shirshendu Das**, "RSP: Restricted Static Pseudo-Partitioning for Mitigation of Cross-Core Covert Channel Attacks," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 29(2), 2023, p. 1-22. **IF: 2, Q2**
6. Atul Kumar, Dipika Deb, **Shirshendu Das**, and Palash Das, "edAttack: Hardware Trojan Attack on On-Chip Packet Compression", *IEEE Design and Test journal - accepted through NOCS-* 40(6), 2023, p. 125-135. **IF: 1.9, Q2**
7. Jaspinder Kaur and **Shirshendu Das**, "TPPD: Targeted Pseudo Partitioning based Defence for Cross-Core Covert Channel Attacks", *Elsevier Journal of System Architecture, Accepted (JSA)*, 135, 2022, p. 102805. **IF: 4.1, Q1**

Journals (Before Joining IITH):

1. Divyansh S. Maura, Tanmay Goel, Kaustav Goswami, Dip Sankar Banerjee, and **Shirshendu Das**, "Variation Aware Power Management for GPU Memories", *Elsevier Journal of Microprocessors and Microsystems (MICPRO)*, 96, 2022, p.104711. **IF: 2.6, Q2**
2. Bharat Bisht, and **Shirshendu Das**, "BHT-NoC: Blaming Hardware Trojans in NoC Routers", *IEEE Design and Test journal, Accepted through NOCS*, 39(6), 2022, p. 39-47. **IF: 1.9, Q2**

3. Sudershan Kumar, Prabuddha Sinha, and **Shirshendu Das**, "WinDRAM: Weak Rows as in-DRAM Cache", *Journal of Concurrency and Computation: Practice and Experience*, 34(28), 2022, p. 7350. **IF: 1.5, Q2**
4. Bindu Agarwalla, **Shirshendu Das** and Nilkanta Sahu, "Process Variation Aware DRAM-Cache Resizing", *Elsevier Journal of System Architecture*, Accepted (JSA), 123, 2021, p. 102364. **IF: 4.1, Q1**
5. Jaspinder Kaur and **Shirshendu Das**, "A Survey on Cache Timing Channel Attacks for Multicore Processors", *Journal of Hardware and Systems Security*, 5(2), 2021, p. 169-189.
6. Kaustav Goswami, Dip Sankar Banerjee and **Shirshendu Das**, "Towards Enhanced System Efficiency While Mitigating Row Hammer", *ACM Transactions on Architecture and Code Optimization (TACO)*, 18(4), 2021, p. 1-26. **IF: 1.8, Q2**
7. Bindu Agarwalla, **Shirshendu Das** and Nilkanta Sahu, "Efficient Cache Resizing Policy for DRAM-based LLCs in ChipMultiprocessors", *Elsevier Journal of System Architecture (JSA)*, 113, 2020, p. 101886. **IF: 4.1, Q1**
8. Dipika Deb, John Jose , **Shirshendu Das** and H. K. Kapoor, "Cost Effective Routing Techniques in 2D Mesh NoC using On-Chip Transmission Lines," *Elsevier Journal of Parallel and Distributed Computing (JPDC)*, 123, 2019, p. 118-129. **IF: 4.0, Q1**
9. **Shirshendu Das** and H. K. Kapoor, "Dynamic Associativity Management in Tiled CMPs by Runtime Adaptation of Fellow Sets," *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 28(8), 2017, p. 2229-2243. **IF: 6, Q1**
10. **Shirshendu Das** and H. K. Kapoor, "A Framework for Block Placement, Migration and Fast Searching in Tiled-DNUCA Architecture," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 22(1), 2016. **IF: 2, Q2**
11. Shirshendu Das and H. K. Kapoor, "Victim Retention for Reducing Cache Misses in Tiled Chip Multiprocessors," *Journal of Microprocessors and Microsystems (Elsevier)*, 38 (4), 2014, p. 263–275. **IF: 2, Q2**
12. **Shirshendu Das**, P. S. Duggirala, and H. K. Kapoor, "A formal framework for interfacing mixed-timing systems," *Integration, the VLSI Journal*, 46 (3), 2013, p. 255-264. **IF:2.5, Q2**
13. H. K. Kapoor, P. Kanakala, M. Verma, and **Shirshendu Das**, "Design and formal verification of a hierarchical cache coherence protocol for noc based multiprocessors," *The Journal of Supercomputing*, 65 (2), 2013, p. 771-796. **IF: 2.7, Q2**

Selected conferences (After Joining IITH):

1. Prathamesh Nitin Tanksale, Guru Raghava S. Seethiraju, **Shirshendu Das** and Venkata Kalyan Tavva, "RRR : Rethinking Randomized Remapping for High Performance Secured NVM LLC", *IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, 2025. **Rank: Highly respected in the hardware security domain.**
2. Aditya S Gangwar, Prathamesh N Thanksale, **Shirshendu Das**, and Sudeepta Mishra, "Flush+earlyReload: Covert Channels Attack on Shared LLC Using MSHR Merging", *Design, Automation and Test in Europe Conference (DATE)* 2024. **Rank: A**
3. Prabuddha Sinha, Krishna Pratik Bv, **Shirshendu Das** and Venkata Kalyan Tavva, "PROLONG: Priority based Write Bypassing Technique for Longer Lifetime in STT-RAM based LLC", *The International Symposium on Memory Systems (MEMSYS)*, 2024.
4. Jaspinder Kaur, and **Shirshendu Das**, "ACPC: Covert Channel Attack on Last Level Cache using Dynamic Cache Partitioning," *24th International Symposium on Quality Electronic Design (ISQED)*, 2023.

Selected Conferences (Before Joining IITH):

1. Kaustav Goswami, **Shirshendu Das**, Sagar Satapathy and Dip Sankar Banerjee, "A Case for Amplifying Row Hammer Attacks via Cell-Coupling in DRAM Devices", *The International Symposium on Memory Systems (MEMSYS)*, 2022.
2. Samiksha Verma and **Shirshendu Das**, "Hybrid Refresh: Improving DRAM Performance by Handling Weak Rows Smartly", *The International Symposium on Memory Systems (MEMSYS)*, 2022.
3. Manaal M. Jamadar, Jaspinder Kaur and **Shirshendu Das**, "MAPCP: Memory Access Pattern Classifying Prefetcher", *The International Symposium on Memory Systems (MEMSYS)*, 2021.
4. Tanmay Goel, Divyansh Maura, Kaustav Goswami, **Shirshendu Das** and Dip Sankar Banerjee, "Towards Row Sensitive DRAM Refresh through Retention Awareness", *ISQED* 2021.

5. Kousik Kumar Dutta, Prathamesh Nitin Tanksale and **Shirshendu Das**, "A Fairness Conscious Cache Replacement Policy for Last Level Cache", *Design, Automation and Test in Europe Conference (DATE)*, 2021. **Rank: A**
6. Anurag Agarwal, Jaspinder Kaur and **Shirshendu Das**, "Exploiting Secrets by Leveraging Dynamic Cache Partitioning of Last Level Cache", *Design, Automation and Test in Europe Conference (DATE)*, 2021. **Rank: A**
7. Kaustav Goswami, Hemanta Kumar Mondal, **Shirshendu Das** and Dip Sankar Banerjee, "State Preserving Dynamic DRAM Bank Re-Configurations for Enhanced Power Efficiency," *International Symposium on Quality Electronic Design (ISQED)*, 2019.
8. **Shirshendu Das** and Hemangee K. Kapoor, "Dynamic Associativity Enabled DNUCA to Improve Block Localisation in Tiled CMPs," *31st ACM/SIGAPP Symposium On Applied Computing (ACM-SAC)*, 2016. **Rank: A**
9. Shounak Chakraborty, **Shirshendu Das** and Hemangee Kapoor, "Static Energy Efficient Cache Reconfiguration for Dynamic NUCA in Tiled CMPs," *31st ACM/SIGAPP Symposium on Applied Computing (ACM-SAC)*, 2016. **Rank: A**
10. **Shirshendu Das** and Hemangee K. Kapoor, "Dynamic Associativity Management Using Utility Based Way-Sharing," *30th ACM/SIGAPP Symposium On Applied Computing (ACM-SAC)*, 2015. **Rank: A**
11. Hemangee Kapoor, **Shirshendu Das** and Shounak Chakraborty, "Static energy reduction by performance linked cache capacity management in Tiled CMPs," *30th ACM/SIGAPP Symposium on Applied Computing (ACM-SAC)*, 2015. **Rank: A**

DECLARATION

Regarding conference rankings, I would like to clarify that the **CORE** ranking system is not an appropriate metric for Computer Architecture conferences. Many top-tier conferences in this field—such as **MICRO**, **DAC**, and **ISCAS**—are not included in the CORE list. This limitation is acknowledged in CORE's official documentation:

<https://portal.core.edu.au/core/media/filesToDisplay/to-be-ranked.pdf>

As stated in that document, the **recommended alternative ranking source** is the **LCC-UMA (SCIE) database**, available at:

<https://scie.lcc.uma.es/>

Based on this recommended site, the conference rankings referred to in my submission are correctly cited.

All the above informations are correct as per my knowledge.

Regards
Shirshendu Das